

YouBike 2.0 站點資料分析

期中專題 — 第 00 組

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1 資料蒐集與整理

1.1 資料來源說明

本報告使用政府資料開放平臺所提供之 YouBike 2.0 臺北市公共自行車即時資訊。資料以 JSON 格式提供，每筆包含站名、行政區、經緯度、總停車格、可借車輛數、可還空位數及更新時間等欄位。

- 下載網址：https://tcgbusfs.blob.core.windows.net/dotapp/youbike/v2/youbike_immediate.json
- 格式：JSON
- 無須申請金鑰

1.2 資料取得

```
library(jsonlite)
library(dplyr)
library(tidyr)
library(stringr)
library(ggplot2)

theme_set(
  theme_minimal(base_size = 11) +
  theme(text = element_text(family = "Noto Sans CJK TC"))
)
knitr::opts_chunk$set(dev = "cairo_pdf")
```

```
url <- "https://tcgbusfs.blob.core.windows.net/dotapp/youbike/v2/youbike_immediate.json"
raw <- fromJSON(url)
df <- as_tibble(raw)
glimpse(df)
```

```
#> Rows: 1,752
#> Columns: 18
#> $ sno                <chr> "500101001", "50010100~
#> $ sna                <chr> "YouBike2.0_捷運科技大樓站", ~
#> $ sarea              <chr> "大安區", "大安區", "大安區", "~
#> $ mday               <chr> "2026-04-14 21:59:04",~
#> $ ar                <chr> "復興南路二段235號前", "復興南路二段~
#> $ sarean            <chr> "Daan Dist.", "Daan Di~
#> $ snaen             <chr> "YouBike2.0_MRT Techno~
#> $ aren              <chr> "No.235 , Sec. 2 , Fuxin~
#> $ act               <chr> "1", "1", "1", "1", "1~
#> $ srcUpdateTime     <chr> "2026-04-14 21:59:52",~
#> $ updateTime        <chr> "2026-04-14 21:59:52",~
#> $ infoTime          <chr> "2026-04-14 21:59:04",~
#> $ infoDate          <chr> "2026-04-14", "2026-04~
#> $ Quantity          <int> 28, 21, 28, 11, 16, 27~
#> $ available_rent_bikes <int> 6, 7, 10, 4, 1, 1, 2, ~
#> $ latitude           <dbl> 25.02605, 25.02565, 25~
```

```
#> $ longitude          <dbl> 121.5436, 121.5436, 12~
#> $ available_return_bikes <int> 22, 13, 17, 7, 15, 26,~
```

1.3 資料清理

```
df_clean <- df |>
  rename(
    total = Quantity,
    sbi    = available_rent_bikes,
    bemp    = available_return_bikes,
    lat     = latitude,
    lng     = longitude
  ) |>
  mutate(
    total = as.integer(total),
    sbi    = as.integer(sbi),
    bemp    = as.integer(bemp),
    act     = as.integer(act),
    lat     = as.numeric(lat),
    lng     = as.numeric(lng),
    mday    = as.POSIXct(mday, format = "%Y-%m-%d %H:%M:%S"),
    ratio   = ifelse(total > 0, sbi / total, NA_real_)
  ) |>
  filter(act == 1, total > 0)

cat(" 清理後筆數:", nrow(df_clean), "\n")
```

```
#> 清理後筆數： 1733
```

```
cat(" 行政區數  :", n_distinct(df_clean$sarea), "\n")
```

```
#> 行政區數    : 13
```

1.4 資料儲存

```
write.csv(df_clean, "data/youbike_clean.csv", row.names = FALSE)
```

2 探索分析

2.1 各行政區站點數量

```
df_clean |>
  count(sarea, sort = TRUE) |>
  mutate(sarea = reorder(sarea, n)) |>
  ggplot(aes(n, sarea)) +
  geom_col(fill = "steelblue") +
  labs(x = " 站點數", y = NULL, title = " 各行政區 YouBike 2.0 站點數量")
```

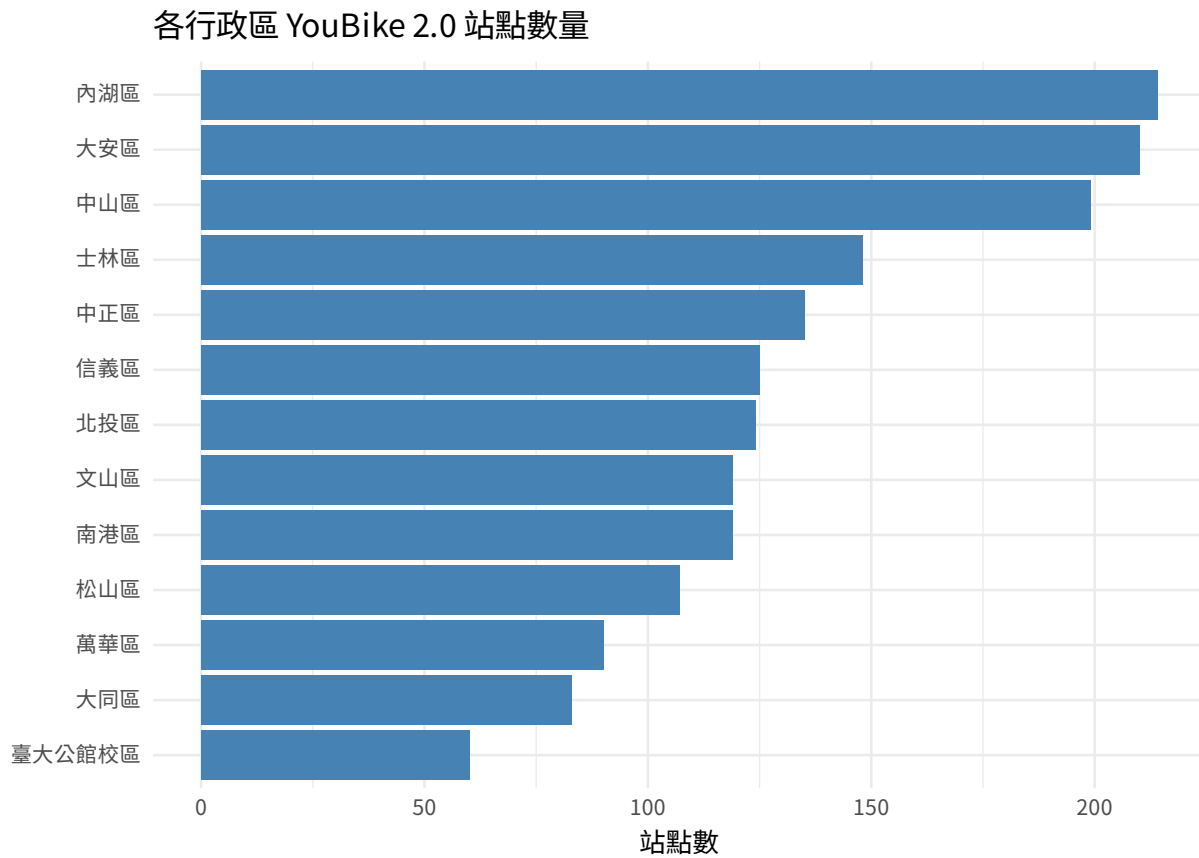


Figure 1: 各行政區 YouBike 2.0 站點數量

Figure 1 顯示各行政區站點數量差異明顯，反映各區人口密度與運輸需求。

2.2 可借車輛比例分布

```
ggplot(df_clean, aes(ratio)) +
  geom_histogram(binwidth = 0.05, fill = "coral", color = "white") +
  scale_x_continuous(labels = scales::percent) +
  labs(x = " 可借車輛 / 總停車格", y = " 站點數",
       title = " 各站可借車輛比例分布")
```

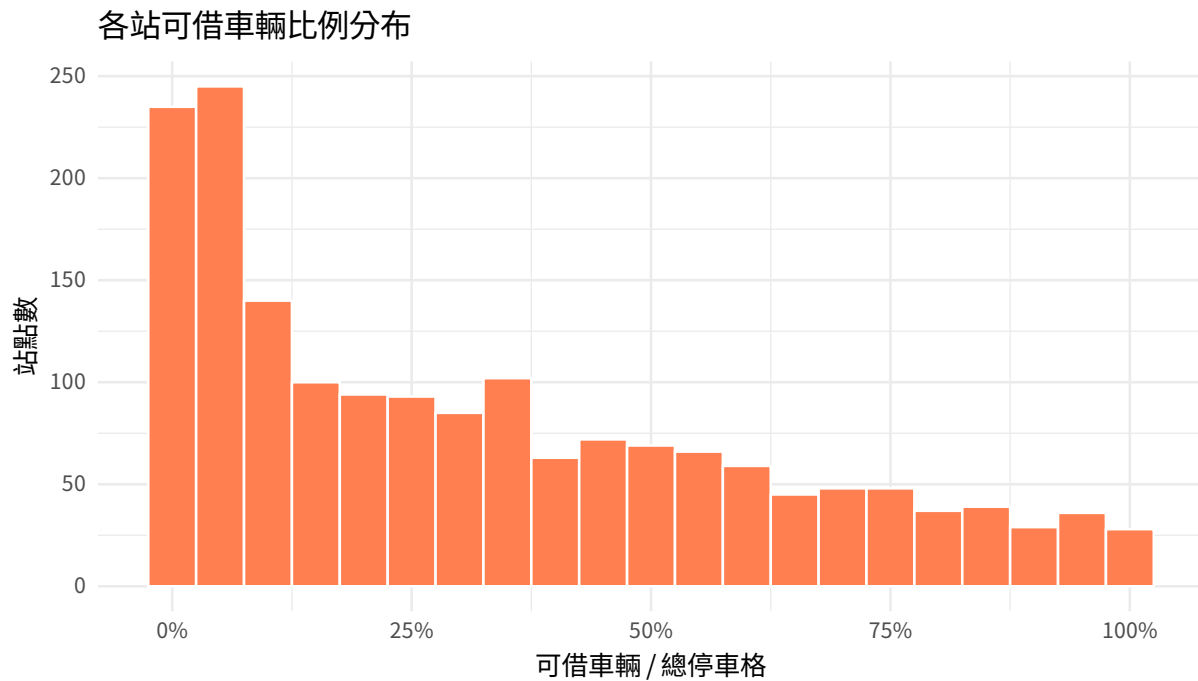


Figure 2: 各站可借車輛比例分布

Figure 2 顯示多數站點車輛比例集中在 20%–60% 之間，兩端極值代表供需失衡站點。

2.3 總停車格 vs 可借車輛

```
top6 <- df_clean |> count(sarea, sort = TRUE) |> slice_head(n = 6) |> pull(sarea)

df_clean |>
  mutate(sarea_label = ifelse(sarea %in% top6, sarea, " 其他")) |>
  ggplot(aes(total, sbi, color = sarea_label)) +
  geom_point(alpha = 0.6, size = 1.5) +
  geom_abline(slope = 1, intercept = 0, linetype = "dashed", color = "gray50") +
  labs(x = " 總停車格", y = " 可借車輛數", color = " 行政區",
       title = " 總停車格 vs 可借車輛") +
  theme(legend.position = "bottom")
```

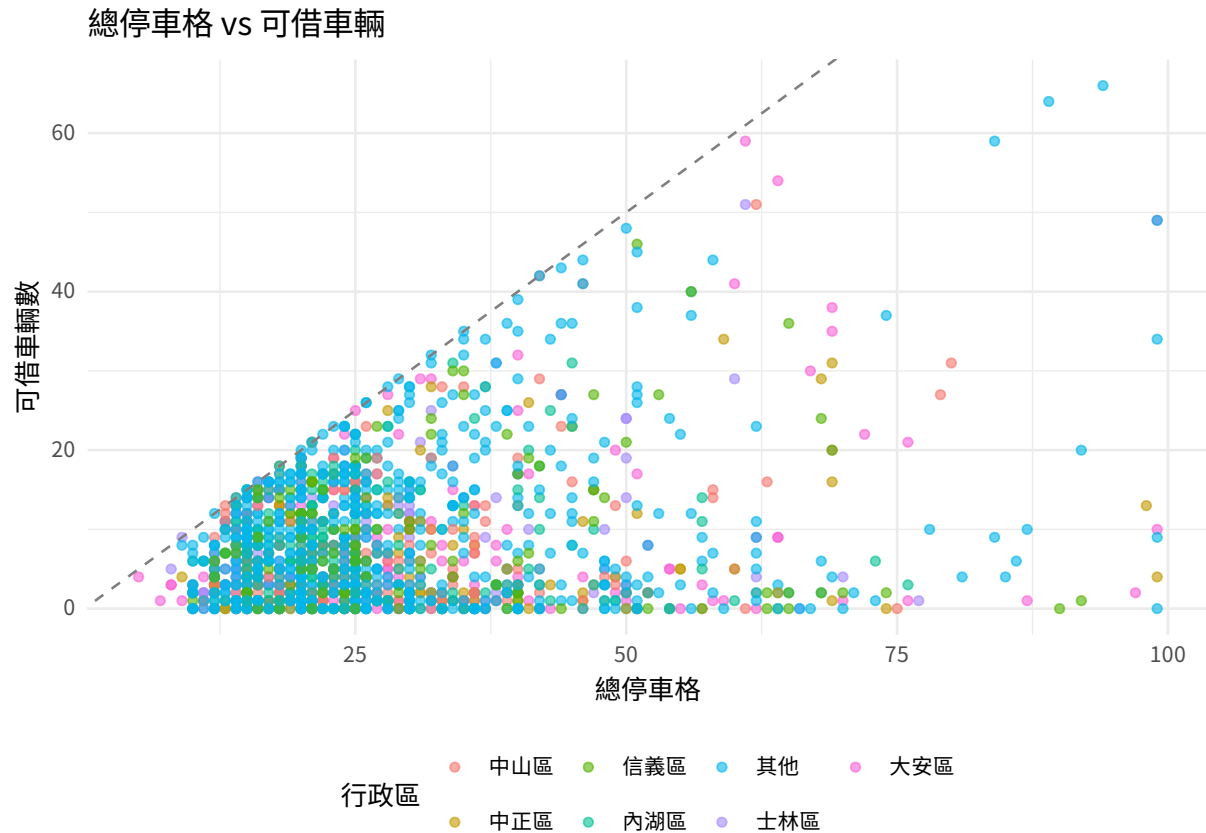


Figure 3: 總停車格 vs 可借車輛 (以行政區著色)

Figure 3 中虛線為 45 度線 (全滿)，大多數站點落在虛線下方，顯示有可還空位。

2.4 供需失衡站點

```
df_clean |>
  slice_min(ratio, n = 10, with_ties = FALSE) |>
  mutate(
    label = str_remove(sna, "^YouBike2\\.0_"),
    label = reorder(label, -ratio)
  ) |>
  select(label, sbi, bemp) |>
  pivot_longer(-label, names_to = "type", values_to = "count") |>
  mutate(type = recode(type, sbi = " 可借車輛", bemp = " 可還空位")) |>
  ggplot(aes(count, label, fill = type)) +
  geom_col(position = "stack") +
  scale_fill_manual(values = c(" 可借車輛" = "steelblue", " 可還空位" = "tomato")) +
  labs(x = " 車位數", y = NULL, fill = NULL,
    title = " 車輛最不足之 10 站") +
  theme(legend.position = "bottom",
    axis.text.y = element_text(size = 7))
```

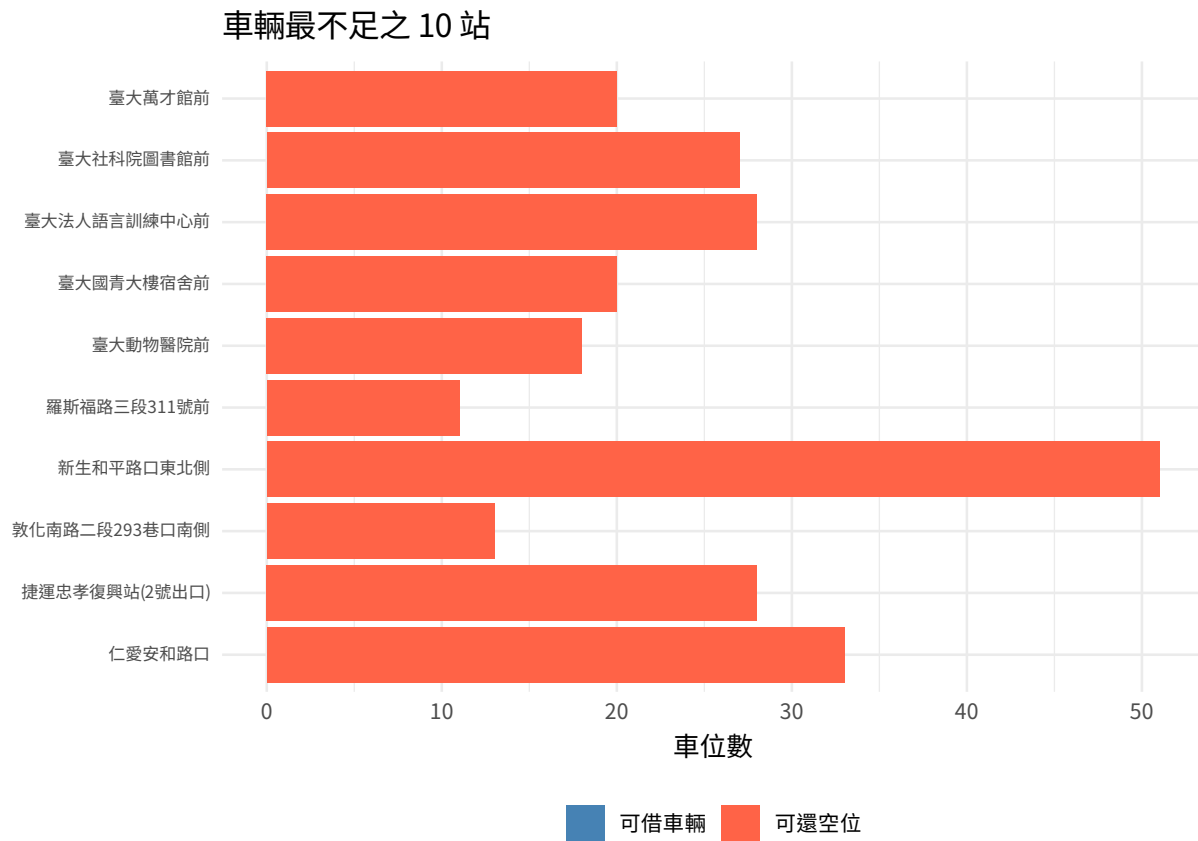


Figure 4: 可借比例最低之 10 站 (可借 vs 可還數量)

Figure 4 列出可借比例最低之 10 站，紅色為空位、藍色為可借車輛，紅色佔比越高代表供需失衡越嚴重。

3 小結

1. 站點集中於人口稠密之行政區 (大安、中山、信義等)
2. 多數站點可借比例介於 20%–60%，少數站點幾乎全滿或全空
3. 總停車格與可借車輛呈正相關，但部分大型站點可借數偏低
4. 識別出若干供需失衡站點，可供調度參考